

林肯教育FIT5196 Week 1 Quiz

总题数: 164

Frank老师整理

按 Mock Quiz 重排（前 6 周）

这份 real quiz 实际覆盖的是前 6 周知识点。我按 `Mock Quiz.pdf` / `mock-quiz.md` 的专题分类重排，并补上了参考答案。
分类优先级：以 mock quiz 的 week 标签为准；因此部分 text pre-processing 题被归到 Week 4，metadata / cataloging / profiling 题被归到 Week 5。

前 6 周 Roadmap

Week 1

- 熟悉 Jupyter Notebook 的 code cell / markdown cell。
- 掌握 Markdown 基础语法：标题、列表、加粗、斜体。
- 建立最基础的 Python/notebook 使用能力。

Week 2

- 理解 Data Wrangling 的目标、挑战与完整流程。
- 能区分 discovery、collection、pre-processing、cleaning、transformation、enrichment、validation、storing。
- 会做基础 pandas 操作，并把具体操作映射回 wrangling task。

Week 3

- 系统学习 regular expressions。
- 掌握 character set、group、anchor、quantifier、lookaround。
- 能写出用于匹配、提取、验证的常见 pattern。

Week 4

- 做 EDA 与 pre-processing：识别数据类型、看分布、看关系。
- 掌握 numerical / categorical / text data 的常见处理。
- 练习 CSV / Excel / JSON 读取，以及 tokenisation、stop words、lemmatisation、vectorisation。

Week 5

- 学习 data discovery 与 data collection 的目标和流程。
- 区分 primary / secondary、structured / unstructured collection method。
- 掌握 metadata、cataloging、profiling、ethics、privacy / security / compliance。

Week 6

- 学习 data structuring。
- 理解 structured / unstructured / graph / time-series 等结构。
- 知道不同结构适合什么场景，以及为什么要先把数据组织好。

逐题分类索引

Week 1 (6 题)

- 范围：Jupyter Notebook、Markdown、Python basics
- [Q1](#) How do you make text italic in Markdown?
- [Q6](#) To create a bullet list in Markdown, which symbol can you use?
- [Q20](#) To emphasise text with bold in Markdown, you can use:
- [Q26](#) For numbered lists in Markdown, what character follows the number? (Select all that apply)
- [Q116](#) To emphasise text with bold in Markdown, you can use:
- [Q155](#) To emphasise text with bold in Markdown, you can use:

Week 2 (48 题)

- 范围：Data Wrangling Process & Tasks + Week 2 pandas applied session
- [Q8](#) How do you select rows 10 to 20 from a DataFrame df? (Select all that apply)
- [Q9](#) Which of the following are considered tasks in the Data Wrangling process? (Select all that apply)
- [Q12](#) Data simplification can involve: (Select all that apply)
- [Q14](#) The goals of data enrichment include: (Select all that apply)
- [Q17](#) Data cleaning may involve: (Select all that apply)
- [Q22](#) How can you rename the column 'A' to 'B' in a DataFrame df?
- [Q33](#) How do you check for missing values in a DataFrame df? (Select all that apply)
- [Q34](#) Scalability in data wrangling is essential for:
- [Q36](#) Data standardization helps ensure data is:
- [Q39](#) What process involves scaling data to a small, specified range?
- [Q41](#) In data quality management, what is crucial for maintaining high-quality data?
- [Q43](#) Data enrichment may involve:
- [Q45](#) Machine Learning enhances data quality management through:
- [Q46](#) Data Integration aims to:
- [Q47](#) Which type of data anomalies refers to a single data point deviating significantly from the rest?
- [Q48](#) Which of the following is NOT a goal of data validation?
- [Q52](#) In data wrangling, data validation ensures: (Select all that apply)
- [Q53](#) How can you rename the column 'A' to 'B' in a DataFrame df?
- [Q59](#) Which of the following are common goals of data wrangling? (Select all that apply)
- [Q67](#) What role does machine learning play in data quality management?
- [Q70](#) Data cleaning may involve: (Select all that apply)
- [Q78](#) Which method would you use to sort a DataFrame df by the column 'Name'?
- [Q79](#) Which is NOT a feature of Big Data?
- [Q80](#) What does simplifying access to data aim to achieve?
- [Q82](#) What is the primary purpose of Data Wrangling?
- [Q83](#) What does data enrichment not involve?
- [Q86](#) The process of identifying and correcting inconsistencies in data is called:

- [Q87](#) Facilitating data integration helps with:
- [Q92](#) Which of the following is NOT a goal of data validation?
- [Q93](#) A major challenge in data wrangling is:
- [Q94](#) Machine Learning enhances data quality management through:
- [Q101](#) Data Integration and Consolidation require: (Select all that apply)
- [Q103](#) How do you select a single column named 'Age' from a DataFrame df? (Select all that apply)
- [Q104](#) Which of the following are common goals of data wrangling? (Select all that apply)
- [Q108](#) In data wrangling, data validation ensures: (Select all that apply)
- [Q109](#) To drop rows with any missing values in the data frame `df`, you use:
- [Q114](#) Data cleaning may involve: (Select all that apply)
- [Q118](#) Which function is used to concatenate two DataFrames vertically?
- [Q119](#) Effective data validation checks are: (Select all that apply)
- [Q121](#) Data integration challenges include:
- [Q125](#) Which type of data anomalies refers to a single data point deviating significantly from the rest?
- [Q126](#) Data duplication is an example of:
- [Q129](#) Which is not a challenge in data wrangling?
- [Q130](#) Which is NOT a challenge in maintaining data quality?
- [Q138](#) The purpose of data aggregation is to:
- [Q143](#) What is a 'Point Anomaly'?
- [Q152](#) Which method would you use to sort a DataFrame df by the column 'Name'?
- [Q160](#) What process involves scaling data to a small, specified range?

Week 3 (39 题)

- 范围: Regular Expressions
- [Q3](#) For matching IPv4 addresses, which regex is accurate and avoids matching numbers beyond the valid range of 0-255?
- [Q7](#) What does a regular expression describe?
- [Q11](#) In the regex `(?:foo)(foo)`, what is the difference between `(?:foo)` and `(foo)`?
- [Q13](#) What does the regex `(?<=\b20)\d{2}\b` match in "Class of 2024"?
- [Q15](#) Which of the following is NOT a character set used in regular expressions?
- [Q18](#) How would you match "http" at the beginning of a string?
- [Q19](#) What will the regex `(?<=\b20)[0-9]{2}\b` match in "Graduating in 2025"?
- [Q21](#) Which character escapes special characters in regular expressions?
- [Q29](#) Which of the following is a valid way to match either 'color' or 'colour'?
- [Q30](#) Given the string 'hello123', what does the regex `(?<=\d)(?=\d)` locate?
- [Q54](#) Which regex finds all instances of "not" that are not immediately followed by "good"?
- [Q56](#) What does the regex `^[\\w.+-]+@([\\w-]+\\.)+[a-zA-Z]{2,7}$` not allow in the email username part?
- [Q58](#) What does the regex `(?<=\b20)\d{2}\b` match in "Class of 2024"?
- [Q61](#) Which metacharacter matches the beginning of a line?
- [Q62](#) How would you match "cat" or "bat"?

- [Q64](#) In regex, what does (?:abc) represent?
- [Q69](#) What does the + quantifier do?
- [Q73](#) What does the {3,6} quantifier mean?
- [Q74](#) In regular expressions, what does \A match?
- [Q76](#) What does the regex `^(?!.*\bfoo\b).*` ensure about the matched strings?
- [Q95](#) Which regex matches a string if it starts with "http://" and does not have "www" immediately after?
- [Q97](#) What does the regex `\A(?:[0-9a-fA-F]{2};){5}[0-9a-fA-F]{2}\z` match?
- [Q99](#) What does the regex `\d{2}(?=\D*$)` match in "Item 23 is 42 years old"?
- [Q100](#) What is a regular expression?
- [Q102](#) What does the regex `\b\d{1,3}(,\d{3})*\b` match?
- [Q106](#) Given the string "hello123", what does the regex `(?<=\D)(?=\d)` locate?
- [Q112](#) What does the regex `(?!cat)dog` ensure?
- [Q113](#) What would the regex `^(\w+)(\s+\1)+$` match?
- [Q115](#) Which character escapes special characters in regular expressions?
- [Q117](#) What is the function of the ? quantifier?
- [Q141](#) What does the * quantifier do in regular expressions?
- [Q142](#) What does `(?<=>)\d+(?=<\d?)` match in a string like "
123
"?
- [Q144](#) Which regex captures text between double quotes, including escaped quotes within the text?
- [Q146](#) What would the regex `\b(?:\d+\b)\w+\b` exclude in its matches?
- [Q147](#) Which regex finds all instances of "not" that are not immediately followed by "good"?
- [Q148](#) Which of the following is NOT a character set used in regular expressions?
- [Q150](#) Which quantifier would match exactly 3 occurrences of the letter 'a'?
- [Q156](#) How would you match any line that does not contain the exact phrase "regex"?
- [Q157](#) What will `(\b[a-zA-Z]{3}\b)\s+\1` match?

Week 4 (41 题)

- 范围：EDA & Pre-processing + Week 4 parsing + mock quiz text pre-processing block
- [Q2](#) What challenges are presented by textual content in EDA for character and text data? (Select all that apply)
- [Q4](#) What factors influence the choice of data pre-processing techniques? (Select all that apply)
- [Q5](#) What measures of central tendency are commonly used in EDA for numerical data? (Select all that apply)
- [Q16](#) Which method is used to get a summary of a DataFrame's structure and column types?
- [Q23](#) For what reasons might case normalisation not always be needed in text data pre-processing? (Select all that apply)
- [Q24](#) For what reasons might case normalisation not always be needed in text data pre-processing? (Select all that apply)
- [Q25](#) The process that simplifies text by converting it to its base form is called:
- [Q27](#) Why is understanding the structure of language (syntax) important in EDA for character and text data? (Select all that apply)
- [Q28](#) Which Pandas function is used to read a CSV file?

- [Q32](#) Which parameter in `pd.read_csv()` specifies the delimiter?
- [Q35](#) What does 'Character Data' consist of?
- [Q37](#) Which technique is used to handle categorical data by creating a binary column for each category?
- [Q40](#) In the context of text data pre-processing, what is the purpose of case normalisation?
- [Q42](#) Exploratory Data Analysis (EDA) is used for:
- [Q55](#) Why is it important to remove stop words in text data pre-processing? (Select all that apply)
- [Q57](#) To read an Excel file into a DataFrame, which function is used?
- [Q63](#) How do you set a DataFrame's column 'ID' as its index?
- [Q66](#) What does the `pd.read_json()` function do?
- [Q71](#) Which techniques are essential for handling numerical data in EDA? (Select all that apply)
- [Q75](#) Which parameter in `read_csv()` is used to specify which rows should be used as header?
- [Q77](#) What does the `skiprows=3` parameter in `pd.read_csv()` do?
- [Q81](#) What is the middle value after all values are ordered in a data set called?
- [Q88](#) Which of the following is crucial for preparing text for NLP tasks?
- [Q90](#) Which measure of central tendency is the arithmetic average of a set of values?
- [Q96](#) For what reasons might case normalisation not always be needed in text data pre-processing? (Select all that apply)
- [Q105](#) Which technique is used to handle categorical data by creating a binary column for each category?
- [Q110](#) Which data types are primarily analysed during EDA? (Select all that apply)
- [Q111](#) Which steps are involved in Lemmatisation for text data pre-processing? (Select all that apply)
- [Q120](#) Which method is used to get a summary of a DataFrame's structure and column types?
- [Q122](#) Which parameter in `pd.read_csv()` specifies the delimiter?
- [Q123](#) What is the aim of using Descriptive Statistics in EDA for Numerical Data?
- [Q124](#) What does continuous data represent?
- [Q131](#) Exploratory Data Analysis (EDA) is used for:
- [Q135](#) Which data visualisation method is primarily used for numerical data to understand its spread and central tendency?
- [Q145](#) Which of the following are benefits of correctly identifying data types during EDA? (Select that all apply)
- [Q149](#) What challenges are presented by textual content in EDA for character and text data? (Select all that apply)
- [Q151](#) What are the steps involved in Count Vectorisation? (Select all that apply)
- [Q153](#) Which of the following are goals of Exploratory Data Analysis (EDA)? (Select all that apply)
- [Q158](#) For what reasons might case normalisation not always be needed in text data pre-processing? (Select all that apply)
- [Q163](#) What is the aim of using Descriptive Statistics in EDA for Numerical Data?
- [Q164](#) What is the significance of identifying data types during EDA?

Week 5 (25 题)

- 范围：Data Discovery & Collection
- [Q10](#) What aspects does Data Cataloging and Metadata Management cover? (Select all that apply)
- [Q31](#) Qualitative Data is:
- [Q44](#) Data Cataloging aids in:

- [Q49](#) Which is NOT a method of collecting structured data?
- [Q50](#) Which of the following is NOT a part of the Data Discovery Process?
- [Q68](#) Factors to consider during the data collection phase include: (Select all that apply)
- [Q72](#) Which are considered primary sources of data? (Select all that apply)
- [Q84](#) Which is NOT a method of collecting structured data?
- [Q85](#) Which phase comes immediately after data discovery?
- [Q89](#) Data Security measures are essential for:
- [Q91](#) Data Cataloging aids in:
- [Q98](#) What roles does data play in decision-making? (Select all that apply)
- [Q107](#) Ethical considerations in Data Collection include: (Select all that apply)
- [Q127](#) Qualitative Data is:
- [Q128](#) The main advantage of using existing data is:
- [Q132](#) Metadata in Data Cataloging includes:
- [Q133](#) What role does data play in modern business decision-making?
- [Q134](#) Metadata Collection is critical for understanding:
- [Q136](#) Ensuring data privacy and security is crucial due to:
- [Q137](#) Which of the following is NOT a part of the Data Discovery Process?
- [Q139](#) Text Mining and Natural Language Processing (NLP) are methods for collecting:
- [Q154](#) What are the roles of Data in modern business decision-making? (Select all that apply)
- [Q159](#) Data Security measures are essential for:
- [Q161](#) Data Privacy measures are intended to:
- [Q162](#) Metadata Collection is critical for understanding:

Week 6 (5 题)

- 范围：Data Structuring
- [Q38](#) Graph data models are ideal for:
- [Q51](#) What does Structured Data refer to?
- [Q60](#) Data Profiling and Assessment involve understanding which types of data structure? (Select all that apply)
- [Q65](#) Which data structures are essential for Data Discovery and Collection? (Select all that apply)
- [Q140](#) Time-series data is indexed in:

歧义题提醒

- Q8：参考按 [a,d](#)；如果老师按 label-based [loc](#) 来解释，也可能接受 [b](#)。
- Q26：参考按 [c](#)；部分 Markdown 方言也接受 [1](#)）。
- Q29：参考按 [c](#)；[d](#) 从 regex 语义上也能匹配。
- Q65：题干把 data structures / sources 混在一起，参考先按 [a,b,c,d,e](#) 记。
- Q95：参考按 [c](#)；[b](#) 也能正确匹配前缀条件。
- Q126：duplicates 既可视作 anomaly，也常被归为 data quality issue；这里按 [b](#)。
- Q144：选项本身不够严谨，参考先按 [c](#)。

原始题库（按索引分类合并）

下方题目顺序已与上面的 Week 1-6 索引完全一致，题号保留原编号。

Week 1 (6 题)

- 范围：Jupyter Notebook、Markdown、Python basics

Question 1

How do you make text italic in Markdown? (Select all that apply)

- a. Surround the text with **
- b. Surround the text with ~~
- c. Surround the text with _
- d. Surround the text with *

参考答案： ☐ c, ☐ d

解析：斜体在 Markdown 里通常写成 `*text*` 或 `_text_`，不是 `**`（粗体）也不是 `~~`（删除线）。

Question 6

To create a bullet list in Markdown, which symbol can you use?

- a. All of the above
- b. *
- c. -
- d. +

参考答案： ☐ a

解析：正确项是 ☐ a: All of the above。它最符合这道题在课程里的定义。

Question 20

To emphasise text with bold in Markdown, you can use:

- a. Double tildes or double dollar signs
- b. Double asterisks or double underscores
- c. Single asterisks or single underscores
- d. Triple asterisks or triple underscores

参考答案： ☐ b

解析：Markdown 粗体用双星号或双下划线，即 `**text**` 或 `__text__`。

Question 26

For numbered lists in Markdown, what character follows the number? (Select all that apply)

- a.)
- b. *
- c. .
- d. -

参考答案: ☒ c

解析: 课程默认的 Markdown 有序列表写法是 1., 所以答案按 ☐ 记。

Question 116

To emphasise text with bold in Markdown, you can use:

- a. Double asterisks or double underscores
- b. Triple asterisks or triple underscores
- c. Single asterisks or single underscores
- d. Double tildes or double dollar signs

参考答案: ☒ a

解析: 和 Q20 同义, 粗体是双星号或双下划线。

Question 155

To emphasise text with bold in Markdown, you can use:

- a. Single asterisks or single underscores
- b. Double tildes or double dollar signs
- c. Triple asterisks or triple underscores
- d. Double asterisks or double underscores

参考答案: ☒ d

解析: 和 Q20/Q116 同义, 粗体仍是双星号或双下划线。

Week 2 (48 题)

- 范围: Data Wrangling Process & Tasks + Week 2 pandas applied session

Question 8

How do you select rows 10 to 20 from a DataFrame df? (Select all that apply)

- a. df.iloc[9:20]
- b. df.loc[10:20]
- c. df.locate[9:20]
- d. df[9:20]

参考答案: ☒ a, ☒ d

解析： `iloc[9:20]` 取的是第 10 到第 20 行（位置索引，右端不含）； `df[9:20]` 也会按切片取对应行。 `loc[10:20]` 依赖标签索引，课内通常不作为标准答案。

Question 9

Which of the following are considered tasks in the Data Wrangling process? (Select all that apply)

- a. Data Modification
- b. Data Storing
- c. Data Validation
- d. Data Presentation
- e. Data Cleaning

参考答案： `b, c, e`

解析： 正确项是 `b`: Data Storing, `c`: Data Validation, `e`: Data Cleaning。这些选项共同对应课程里的正确定义。

Question 12

Data simplification can involve: (Select all that apply)

- a. Enhancing understandability of data
- b. Removing relevant information
- c. Filtering out irrelevant information
- d. Summarizing complex data

参考答案： `a, c, d`

解析： 正确项是 `a`: Enhancing understandability of data, `c`: Filtering out irrelevant information, `d`: Summarizing complex data。这些选项共同对应课程里的正确定义。

Question 14

The goals of data enrichment include: (Select all that apply)

- a. Adding value to the data
- b. Improving data quality
- c. Enhancing data analysis capabilities
- d. Simplifying data formats

参考答案： `a, b, c`

解析： data enrichment 的目标是加上下文、提升价值、改善可分析性；“simplifying data formats”更像 pre-processing / transformation。

Question 17

Data cleaning may involve: (Select all that apply)

- a. Introducing new errors
- b. Removing duplicates
- c. Handling missing values
- d. Correcting errors

参考答案: `b, c, d`

解析: 正确项是 `b`: Removing duplicates, `c`: Handling missing values, `d`: Correcting errors。这些选项共同对应课程里的正确定义。

Question 22

How can you rename the column 'A' to 'B' in a DataFrame df?

- a. `df.columns['A'] = 'B'`
- b. `df.columns.rename('A', 'B')`
- c. `df.rename('A', 'B')`
- d. `df.rename(columns={'A': 'B'})`

参考答案: `d`

解析: Pandas 改列名标准写法是 `df.rename(columns={"A": "B"})`。

Question 33

How do you check for missing values in a DataFrame df? (Select all that apply)

- a. `df.isna()`
- b. `df.has_null()`
- c. `df.missing()`
- d. `df.isnull()`

参考答案: `a, d`

解析: Pandas 检查缺失值常用 `isna()` 和 `isnull()`。

Question 34

Scalability in data wrangling is essential for:

- a. Handling increased data loads without performance loss
- b. Reducing data processing power
- c. Maintaining a constant volume of data
- d. Complicating data integration processes

参考答案: `a`

解析: scalability 指数据量上来后仍能处理, 不是把数据量固定不变。

Question 36

Data standardization helps ensure data is:

- a. In a consistent format
- b. Only available in one unit of measurement
- c. Inconsistent across datasets
- d. Difficult to analyze

参考答案: ☒ a

解析: 这里的 standardization 更偏“统一标准/格式”, 目标是让数据表示保持一致。

Question 39

What process involves scaling data to a small, specified range?

- a. Diversification
- b. Normalisation
- c. Aggregation
- d. Segmentation

参考答案: ☒ b

解析: 把数值缩到一个小范围 (如 `[0,1]`) 是 normalisation。

Question 41

In data quality management, what is crucial for maintaining high-quality data?

- a. Aesthetic appeal
- b. Data volume
- c. Data Governance
- d. Data variety

参考答案: ☒ c

解析: 持续高质量数据离不开 data governance, 因为要靠规则、责任和流程来维护。

Question 43

Data enrichment may involve:

- a. Verifying data accuracy
- b. Adding time-related data
- c. Correcting data errors
- d. Converting data into a binary format

参考答案: ☒ b

解析: enrichment 常见做法是补充时间、地理、人口等额外上下文。

Question 45

Machine Learning enhances data quality management through:

- a. Solely focusing on data visualization
- b. A combination of techniques including NLP and continuous monitoring
- c. Only data cleansing
- d. Only automated error detection

参考答案: ☒ b

解析: 机器学习在 data quality management 里常用于自动识别问题、持续监控和 NLP 等组合方法。

Question 46

Data Integration aims to:

- a. Reduce the overall data quality
- b. Create inconsistent datasets
- c. Combine multiple data sources for a unified view
- d. Provide a disjointed view of information

参考答案: ☒ c

解析: data integration 的目标是把多个来源统一成一个一致视图。

Question 47

Which type of data anomalies refers to a single data point deviating significantly from the rest?

- a. Complex Anomalies
- b. Contextual Anomalies
- c. Collective Anomalies
- d. Point Anomalies

参考答案: ☒ d

解析: 单个点明显偏离整体, 就是 point anomaly。

Question 48

Which of the following is NOT a goal of data validation?

- a. Verify data is in the expected format
- b. Validate range and constraints
- c. Ensure data accurately reflects real-world entities
- d. Enhance the speed of data retrieval

参考答案: ☒ d

解析：validation 是检查格式、范围、逻辑是否合理，不是提升读取速度。

Question 52

In data wrangling, data validation ensures: (Select all that apply)

- a. Data inconsistency is promoted
- b. Data meets predefined standards
- c. Data is in a usable format for analysis
- d. Accuracy and reliability of data

参考答案： `b, c, d`

解析：validation 要保证数据符合预设标准、可靠且可用于后续分析。

Question 53

How can you rename the column 'A' to 'B' in a DataFrame df?

- a. `df.rename(columns={'A': 'B'})`
- b. `df.columns['A'] = 'B'`
- c. `df.rename('A', 'B')`
- d. `df.columns.rename('A', 'B')`

参考答案： `a`

解析：这题和 Q22 同义，标准写法仍是 `df.rename(columns={"A": "B"})`。

Question 59

Which of the following are common goals of data wrangling? (Select all that apply)

- a. Increasing data complexity
- b. Ignoring data inconsistencies
- c. Improving data quality
- d. Facilitating easier access to data

参考答案： `c, d`

解析：wrangling 的常见目标是提高质量、提升可访问性，不是增加复杂度。

Question 67

What role does machine learning play in data quality management?

- a. Enhancing processes of identifying and correcting data quality issues
- b. Reducing the need for data governance
- c. Only for data visualization
- d. Only for database management

参考答案: ☒ a

解析: 机器学习的作用是帮助发现、识别并修正数据质量问题。

Question 70

Data cleaning may involve: (Select all that apply)

- a. Correcting errors
- b. Removing duplicates
- c. Handling missing values
- d. Introducing new errors

参考答案: ☒ a, ☒ b, ☒ c

解析: cleaning 包括纠错、去重、处理缺失, 不会“引入新错误”。

Question 78

Which method would you use to sort a DataFrame df by the column 'Name'?

- a. `df.rank('Name')`
- b. `df.sort_values(by='Name')`
- c. `df.sort_by('Name')`
- d. `df.sort('Name')`

参考答案: ☒ b

解析: 按列排序标准写法是 `df.sort_values(by="Name")`。

Question 79

Which is NOT a feature of Big Data?

- a. Manageable by traditional software
- b. Voluminous
- c. Includes various data structures
- d. Involves high velocity

参考答案: ☒ a

解析: Big Data 的典型特征是 volume、variety、velocity, 通常不能被传统软件轻松处理。

Question 80

What does simplifying access to data aim to achieve?

- a. Increase the complexity of data access
- b. Organise data in a user-friendly manner
- c. Make data accessible only to data scientists
- d. Restrict data access

参考答案: ☒ b

解析: simplifying access 的目标是让数据更容易找、更容易用。

Question 82

What is the primary purpose of Data Wrangling?

- a. To analyze complex data types
- b. To ensure data privacy
- c. To acquire, clean, structure, and enrich raw data for analysis
- d. To visualize data trends

参考答案: ☒ c

解析: data wrangling 的核心就是把原始数据整理成可分析数据。

Question 83

What does data enrichment not involve?

- a. Creating new variables
- b. Reducing data insights
- c. Adding external context
- d. Enhancing data with additional information

参考答案: ☒ b

解析: enrichment 是增加信息与洞察, 不是减少 insights。

Question 86

The process of identifying and correcting inconsistencies in data is called:

- a. Data collection
- b. Data integration
- c. Data cleaning
- d. Data enrichment

参考答案: ☒ c

解析: 识别并修正不一致, 属于 data cleaning。

Question 87

Facilitating data integration helps with:

- a. Ignoring data quality
- b. Keeping data in isolated systems
- c. Transforming disparate data into a common format
- d. Making it harder to combine data from different sources

参考答案: ☒ c

解析: integration 的一个关键作用就是把异构数据转成共同格式。

Question 92

Which of the following is NOT a goal of data validation?

- a. Validate range and constraints
- b. Ensure data accurately reflects real-world entities
- c. Enhance the speed of data retrieval
- d. Verify data is in the expected format

参考答案: ☒ c

解析: 和 Q48 同理, validation 不以提升检索速度为目标。

Question 93

A major challenge in data wrangling is:

- a. Avoiding data standardization
- b. Maintaining low data quality
- c. Data privacy and security
- d. Handling small volumes of data

参考答案: ☒ c

解析: data privacy and security 是 data wrangling 的典型难点之一。

Question 94

Machine Learning enhances data quality management through:

- a. Solely focusing on data visualization
- b. A combination of techniques including NLP and continuous monitoring
- c. Only data cleansing
- d. Only automated error detection

参考答案: ☒ b

解析： 这题和 Q45 同义，关键是 ML 能结合多种自动化方法提升质量管理。

Question 101

Data Integration and Consolidation require: (Select all that apply)

- a. Combining data sources
- b. Harmonise data formats
- c. Data anonymisation
- d. Data cleaning
- e. Ensuring consistency

参考答案： `a, b, d, e`

解析： integration / consolidation 要合并来源、统一格式、清洗并保证一致性。

Question 103

How do you select a single column named 'Age' from a DataFrame df? (Select al that apply)

- a. `df['Age']`
- b. `df.select('Age')`
- c. `df.index('Age')`
- d. `df.Age`

参考答案： `a, d`

解析： 单列可用 `df["Age"]` 或属性访问 `df.Age`。

Question 104

Which of the following are common goals of data wrangling? (Select all that apply)

- a. Increasing data complexity
- b. Ignoring data inconsistencies
- c. Facilitating easier access to data
- d. Improving data quality

参考答案： `c, d`

解析： 和 Q59 同义，目标仍是提升质量与可访问性。

Question 108

In data wrangling, data validation ensures: (Select all that apply)

- a. Data meets predefined standards
- b. Accuracy and reliability of data
- c. Data inconsistency is promoted
- d. Data is in a usable format for analysis

参考答案: `a, b, d`

解析: 和 Q52 同义, validation 关注标准、可靠性和可用性。

Question 109

To drop rows with any missing values in the data frame `df`, you use:

- a. `df.fillna()`
- b. `df.dropna()`
- c. `df.remove_na()`
- d. `df.delete_missing()`

参考答案: `b`

解析: 删除含缺失值的行用 `df.dropna()`。

Question 114

Data cleaning may involve: (Select all that apply)

- a. Removing duplicates
- b. Correcting errors
- c. Handling missing values
- d. Introducing new errors

参考答案: `a, b, c`

解析: 和 Q17/Q70 同义, cleaning 还是去重、纠错、处理缺失。

Question 118

Which function is used to concatenate two DataFrames vertically?

- a. `pd.concat()`
- b. `pd.join()`
- c. `pd.merge()`
- d. `pd.append()`

参考答案: `a`

解析: 纵向拼接两个 DataFrame 通常用 `pd.concat()`。

Question 119

Effective data validation checks are: (Select all that apply)

- a. Completeness checking
- b. Range checking
- c. Encryption methods
- d. Consistency checking

参考答案: ☒ a, ☒ b, ☒ d

解析: 有效 validation check 常见有 completeness、range、consistency; encryption 不是 validation check。

Question 121

Data integration challenges include:

- a. Combining data of varying formats and quality
- b. Maintaining data in disparate formats
- c. Simplifying the combination of similar data
- d. Ignoring data from different sources

参考答案: ☒ a

解析: integration 真正难在格式和质量不一致, 不是“忽略不同来源”。

Question 125

Which type of data anomalies refers to a single data point deviating significantly from the rest?

- a. Collective Anomalies
- b. Point Anomalies
- c. Complex Anomalies
- d. Contextual Anomalies

参考答案: ☒ b

解析: 和 Q47 同义, 这里 point anomaly 的选项顺序变了。

Question 126

Data duplication is an example of:

- a. Regular expression pattern
- b. Data quality issue
- c. Data wrangling task
- d. Data anomaly

参考答案: ☒ b

解析: 重复数据首先是 data quality issue; 虽然也常被看作 anomaly, 但复习时先按课程答案记。

Question 129

Which is not a challenge in data wrangling?

- a. Handling large volumes of data
- b. Ensuring data privacy and security
- c. Maintaining outdated data formats
- d. Decreasing analytical efficiency

参考答案: ☒ d

解析: 前三项都是挑战; “decreasing analytical efficiency” 不是挑战表述, 而是负面结果说法。

Question 130

Which is NOT a challenge in maintaining data quality?

- a. Complexity of data integration
- b. Volume and variety of data
- c. Data silos
- d. High-quality data tools

参考答案: ☒ d

解析: 真正的挑战是集成复杂、量大、数据孤岛; 高质量工具本身不是挑战。

Question 138

The purpose of data aggregation is to:

- a. Increase data detail
- b. Disperse data across sources
- c. Summarise data for analysis
- d. Complicate data analysis

参考答案: ☒ c

解析: aggregation 的目的就是把明细汇总成更适合分析的结果。

Question 143

What is a 'Point Anomaly'?

- a. A data point that adheres perfectly to the data set pattern
- b. A data point that significantly deviates from the rest of the data set
- c. A standard deviation in statistical analysis
- d. A group of data points that are not unusual

参考答案: ☒ b

解析： point anomaly 就是单个点显著偏离整体。

Question 152

Which method would you use to sort a DataFrame df by the column 'Name'?

- a. `df.sort_values(by='Name')`
- b. `df.sort('Name')`
- c. `df.rank('Name')`
- d. `df.sort_by('Name')`

参考答案： ☐ a

解析： 和 Q78 同义，只是正确项顺序换成了 ☐ a。

Question 160

What process involves scaling data to a small, specified range?

- a. Diversification
- b. Aggregation
- c. Normalisation
- d. Segmentation

参考答案： ☐ c

解析： 和 Q39 同义，缩放到小范围依然是 normalisation。

Week 3 (39 题)

- 范围： Regular Expressions

Question 3

For matching IPv4 addresses, which regex is accurate and avoids matching numbers beyond the valid range of 0-255?

- a. `\b(0-255){1,3}.(0-255){1,3}.(0-255){1,3}.(0-255){1,3}\b`
- b. `[0-9]{1,3}.[0-9]{1,3}{3}`
- c. `\b(25[0-5] | 2[0-4][0-9] | [01]?[0-9][0-9]?).(25[0-5] | 2[0-4][0-9] | [01]?[0-9][0-9]?).(25[0-5] | 2[0-4][0-9] | [01]?[0-9][0-9]?).(25[0-5] | 2[0-4][0-9] | [01]?[0-9][0-9]?)\b`
- d. `\b(\d{1,3}).{3}\d{1,3}\b`

参考答案： ☐ c

解析： 只有 ☐ c 把每一段都限制在 `0-255` 范围内； ☐ b 和 ☐ d 只限制位数，不限制数值大小。

Question 7

What does a regular expression describe?

- a. Text pattern
- b. Image pattern
- c. Numeric calculation
- d. Audio frequency

参考答案: ☒ a

解析: 正确项是 ☒ a: Text pattern。它最符合这道题在课程里的定义。

Question 11

In the regex `(?:foo)(foo)`, what is the difference between `(?:foo)` and `(foo)`?

- a. `(?:foo)` matches and captures "foo"; `(foo)` only matches "foo"
- b. Neither captures "foo"; they only match
- c. `(?:foo)` only matches "foo"; `(foo)` matches and captures "foo"
- d. Both capture "foo" for future use

参考答案: ☒ c

解析: `(?:foo)` 是非捕获组, 只分组不保存; `(foo)` 是捕获组, 会把匹配结果保存下来供后续引用。

Question 13

What does the regex `(?<=\b20)\d{2}\b` match in "Class of 2024"?

- a. 2024
- b. Class of
- c. 24
- d. 20

参考答案: ☒ c

解析: lookbehind `(?<=\b20)` 要求前面紧跟 20, 因此真正被匹配的是后面的两位数字 24。

Question 15

Which of the following is NOT a character set used in regular expressions?

- a. `[a-zA-Z]`
- b. `\d`
- c. `[0-9]`
- d. `(a-z)`

参考答案: ☒ d

解析: 正确项是 ☒ d: (a-z)。它最符合这道题在课程里的定义。

Question 18

How would you match "http" at the beginning of a string?

- a. \bhttp
- b. http\$
- c. http\b
- d. ^http

参考答案: ☒ d

解析: ☐ ^ 匹配字符串开头, 所以要写 ☒ ^http。

Question 19

What will the regex `(?<=\b20)[0-9]{2}\b` match in "Graduating in 2025"?

- a. 2025
- b. 25
- c. 20
- d. No match

参考答案: ☒ b

解析: 和 Q13 同理, `(?<=\b20)` 只把 `20` 后面的两位数取出来, 所以是 `25`。

Question 21

Which character escapes special characters in regular expressions?

- a. ^
- b. \$
- c. |
- d. \

参考答案: ☒ d

解析: 正确项是 ☒ d: \。它最符合这道题在课程里的定义。

Question 29

Which of the following is a valid way to match either 'color' or 'colour'?

- a. colo(u|r)
- b. colou|r
- c. colou?r
- d. (color|colour)

参考答案: ☒ c

解析: `colou?r` 中 `u?` 表示 `u` 可有可无, 因此能同时匹配 `color` 和 `colour`。`(color|colour)` 也能匹配, 但课里通常把 `c` 当标准答案。

Question 30

Given the string 'hello123', what does the regex `(?<=\D)(?=\d)` locate?

- a. The position between 'o' and '1'
- b. The sequence "123"
- c. The digit "1"
- d. The letter 'o'

参考答案: `a`

解析: 这个零宽断言定位的是“非数字后面紧跟数字”的那个位置, 也就是 `o` 和 `1` 中间。

Question 54

Which regex finds all instances of "not" that are not immediately followed by "good"?

- a. `not(?: good)`
- b. `not(?:= good)`
- c. `not(?:=good)`
- d. `not(?:!good)`

参考答案: `d`

解析: 负向前瞻 `(?!good)` 表示后面不能立刻跟 `good`, 所以选 `not(?:!good)`。

Question 56

What does the regex `^[\\w.+~]+@[\\w-]+\\.([a-zA-Z]{2,7})$` not allow in the email username part?

- a. Spaces
- b. Underscore
- c. Plus signs
- d. Digits

参考答案: `a`

解析: 用户名部分的字符类里不包含空格, 所以空格不允许。

Question 58

What does the regex `(?<=\\b20)\\d{2}\\b` match in "Class of 2024"?

- a. 20
- b. 24
- c. 2024
- d. Class of

参考答案: **b**

解析: 和 Q13/Q19 同理, 真正被匹配到的是 **20** 后面的两位, 即 **24**。

Question 61

Which metacharacter matches the beginning of a line?

- a. ^
- b. |
- c. \b
- d. \$

参考答案: **a**

解析: 行首锚点是 **^**。

Question 62

How would you match "cat" or "bat"?

- a. [cb]at
- b. cat|bat
- c. (cb)at
- d. [c&b]at

参考答案: **a**

解析: **[cb]at** 表示首字母可以是 **c** 或 **b**。

Question 64

In regex, what does (?:abc) represent?

- a. An optional character sequence
- b. A non-capturing group
- c. A capturing group
- d. A repeated sequence

参考答案: **b**

解析: **(?:abc)** 是 non-capturing group。

Question 69

What does the + quantifier do?

- a. Matches exactly one occurrence
- b. Matches zero or more occurrences
- c. Matches zero or one occurrence
- d. Matches one or more occurrences

参考答案: ☐ d

解析: ☐ + 表示前一项出现 1 次或多次。

Question 73

What does the {3,6} quantifier mean?

- a. Matches at least 3 occurrences of the preceding element
- b. Matches up to 6 occurrences of the preceding element
- c. Matches between 3 and 6 occurrences of the preceding element
- d. Matches exactly 3 or exactly 6 occurrences of the preceding element

参考答案: ☐ c

解析: ☐ {3,6} 就是前一项出现 3 到 6 次。

Question 74

In regular expressions, what does \A match?

- a. The beginning of a string
- b. A word boundary
- c. The end of a string
- d. A non-word boundary

参考答案: ☐ a

解析: ☐ \A 匹配整个字符串开头。

Question 76

What does the regex ☐ `^(?!.*\bfoo\b).*$` ensure about the matched strings?

- a. They must not contain "foo"
- b. They must contain "foo"
- c. They start with "foo"
- d. They end with "foo"

参考答案: ☐ a

解析: 整条表达式用负向前瞻排除了包含独立单词 ☐ foo 的字符串。

Question 95

Which regex matches a string if it starts with "http://" and does not have "www" immediately after?

- a. `^http(?:!://www.)`
- b. `^http:\V(?:!www.)`
- c. `^http:\V(?:!www).*`
- d. `^https?(?:!://www.)`

参考答案: ☒ c

解析: `^http://(?:!www).*` 的思路最直接: 以 `http://` 开头, 且后面不是 `www`。`b` 也能匹配, 但这里先按 `c` 记。

Question 97

What does the regex `\A(?:[0-9a-fA-F]{2}:){5}[0-9a-fA-F]{2}\z` match?

- a. An IPv6 address
- b. A MAC address
- c. An IPv4 address
- d. An email address

参考答案: ☒ b

解析: 六组两位十六进制加冒号分隔, 是典型 MAC address。

Question 99

What does the regex `\d{2,}(?=\D*$)` match in "Item 23 is 42 years old"?

- a. 2342
- b. 23
- c. No match
- d. 42

参考答案: ☒ d

解析: 正则要找的是“最后一个连续数字片段”, 所以会匹配到句尾前的 `42`。

Question 100

What is a regular expression?

- a. A way to calculate expressions in math
- b. A method for searching text in a document
- c. A sequence of characters that define a search pattern
- d. A programming language

参考答案: ☒ c

解析： regular expression 本质上就是定义搜索模式的一串字符。

Question 102

What does the regex `\b\d{1,3}(\,\d{3})*\b` match?

- a. A number formatted with commas (e.g., "1,000")
- b. A comma-separated list of numbers
- c. An IP address
- d. Any digit sequence without commas

参考答案： ☐ a

解析： 这个模式匹配像 `1,000`、`12,345,678` 这样的千分位数字格式。

Question 106

Given the string "hello123", what does the regex `(?<=\D)(?=\d)` locate?

- a. The digit "1"
- b. The letter "o"
- c. The position between "o" and "1"
- d. The sequence "123"

参考答案： ☐ c

解析： 和 Q30 同义，定位的是 `o` 与 `1` 之间的位置。

Question 112

What does the regex `(?!cat)dog` ensure?

- a. "dog" is preceded by "cat"
- b. "dog" is not preceded by "cat"
- c. "dog" follows "cat"
- d. Both "cat" and "dog" are present

参考答案： ☐ b

解析： 负向后顾 `(?!cat)` 要求 `dog` 前面不能是 `cat`。

Question 113

What would the regex `^(w+)(s+\1)+$` match?

- a. A word repeated at least once, separated by whitespace
- b. Any number of repeated words without spaces
- c. A single word with trailing whitespace
- d. A line containing only whitespace

参考答案: ☐ a

解析: 这个表达式通过反向引用 `\1` 匹配“同一个词重复出现”。

Question 115

Which character escapes special characters in regular expressions?

- a. ^
- b. |
- c. \
- d. \$

参考答案: ☐ c

解析: 和 Q21 同义, 转义特殊字符靠反斜杠。

Question 117

What is the function of the `?` quantifier?

- a. Matches one or more occurrences
- b. Matches zero or more occurrences
- c. Matches zero or one occurrence
- d. Matches exactly one occurrence

参考答案: ☐ c

解析: `?` 表示前一项出现 0 次或 1 次。

Question 141

What does the `*` quantifier do in regular expressions?

- a. Matches exactly one character
- b. Matches zero or more occurrences of the preceding element
- c. Matches one or more occurrences of the preceding element
- d. Matches zero or one occurrence of the preceding element

参考答案: ☐ b

解析: `*` 表示前一项出现 0 次或多次。

Question 142

What does `(?<=>)\d+(?=<\/?)` match in a string like "

123

"?

- a. The entire string including the tags
- b. Any digits followed by "</"
- c. Any digits preceded by ">"
- d. 123, if it appears inside HTML tags

参考答案: ☐ d

解析: 这个模式会把 HTML 标签之间的数字内容 123 抠出来。

Question 144

Which regex captures text between double quotes, including escaped quotes within the text?

- a. `"[^\"]+"`
- b. `\\\"[\\w\\s]*\\\"`
- c. `"(.*)"`
- d. `".*"`

参考答案: ☐ c

解析: 在给出的选项里, `"(.*)"` 最接近“抓取双引号之间内容”的写法; 但这题选项本身不够严谨。

Question 146

What would the regex `\b(?:\d+\b)\w+\b` exclude in its matches?

- a. Words ending with a digit
- b. Words starting with a digit
- c. Words containing any digits
- d. Words entirely composed of digits

参考答案: ☐ d

解析: 负向前瞻 `(?:\d+\b)` 排除了“整个词就是纯数字”的情况。

Question 147

Which regex finds all instances of "not" that are not immediately followed by "good"?

- a. `not(?:!good)`
- b. `not(?:=good)`
- c. `not(?:! good)`
- d. `not(?:= good)`

参考答案: ☐ a

解析：和 Q54 同义，这里标准答案写成了带反引号的 `not(?!good)`。

Question 148

Which of the following is NOT a character set used in regular expressions?

- a. `\d`
- b. `[a-zA-Z]`
- c. `[0-9]`
- d. `{a-z}`

参考答案： `d`

解析： `{a-z}` 不是字符集写法；字符集要用方括号。

Question 150

Which quantifier would match exactly 3 occurrences of the letter 'a'?

- a. `a3`
- b. `a{3}`
- c. `a[3]`
- d. `a(3)`

参考答案： `b`

解析：恰好 3 次要用 `{3}`，所以是 `a{3}`。

Question 156

How would you match any line that does not contain the exact phrase "regex"?

- a. `^(?=\bregex\b).$`
- b. `^(?!.\bregex\b).`
- c. `.\bregex\b.`
- d. `^(regex).`

参考答案： `b`

解析：负向前瞻 `^(?!.*\bregex\b).*` 用来排除包含 `regex` 的整行。

Question 157

What will `(\b[a-zA-Z]{3}\b)s+\1` match?

- a. Any word repeated exactly once after one or more spaces
- b. A three-letter word followed by another three-letter word
- c. Any three-letter word repeated with one or more spaces in between
- d. A sequence of six letters with a space in the middle

参考答案: ☒ c

解析: 这里要求“三个字母的单词重复出现”，且中间有空格。

Week 4 (41 题)

- 范围: EDA & Pre-processing + Week 4 parsing + mock quiz text pre-processing block

Question 2

What challenges are presented by textual content in EDA for character and text data? (Select all that apply)

- a. Variability in text formats
- b. Data sparsity
- c. Ambiguity in language
- d. High dimensionality
- e. High computational costs

参考答案: ☒ a, ☒ b, ☒ c, ☒ d, ☒ e

解析: 正确项是 ☒ a: Variability in text formats, ☒ b: Data sparsity, ☒ c: Ambiguity in language, ☒ d: High dimensionality, ☒ e: High computational costs。这些选项共同对应课程里的正确定义。

Question 4

What factors influence the choice of data pre-processing techniques? (Select all that apply)

- a. Algorithm requirements
- b. Data quantity
- c. Data integrity
- d. Visualisation needs
- e. Data type

参考答案: ☒ a, ☒ b, ☒ c, ☒ d, ☒ e

解析: 正确项是 ☒ a: Algorithm requirements, ☒ b: Data quantity, ☒ c: Data integrity, ☒ d: Visualisation needs, ☒ e: Data type。这些选项共同对应课程里的正确定义。

Question 5

What measures of central tendency are commonly used in EDA for numerical data? (Select all that apply)

- a. Median
- b. Variance
- c. Mode
- d. Standard Deviation
- e. Mean

参考答案: `a, c, e`

解析: 中心趋势看的是数据“中心”在哪里, 常见是 mean / median / mode; variance 和 standard deviation 属于离散程度。

Question 16

Which method is used to get a summary of a DataFrame's structure and column types?

- a. `df.info()`
- b. `df.describe()`
- c. `df.overview()`
- d. `df.summary()`

参考答案: `a`

解析: `df.info()` 会给出列名、非空数、`dtype` 等结构信息; `describe()` 更偏统计摘要。

Question 23

For what reasons might case normalisation not always be needed in text data pre-processing? (Select all that apply)

- a. To maintain the original meaning of words
- b. In information retrieval tasks
- c. For graphical data representation
- d. When performing named entity recognition
- e. In audio data processing

参考答案: `a, b, d`

解析: case normalisation 并不总是必须; 像 named entity recognition 或需要保留原大小写信息时, 大小写本身就有语义。

Question 24

For what reasons might case normalisation not always be needed in text data pre-processing? (Select all that apply)

- a. To maintain the original meaning of words
- b. In information retrieval tasks
- c. For graphical data representation
- d. When performing named entity recognition
- e. In audio data processing

参考答案: `a, b, d`

解析： 这题和 Q23 同义，核心也是“某些任务里大小写信息本身有用，所以不一定统一转小写”。

Question 25

The process that simplifies text by converting it to its base form is called:

- a. Case normalization
- b. One-hot encoding
- c. Tokenisation
- d. Lemmatisation

参考答案： ☒ d

解析： 把词还原成词典中的基本形式是 lemmatisation。

Question 27

Why is understanding the structure of language (syntax) important in EDA for character and text data? (Select all that apply)

- a. It is crucial for visual data representation
- b. It supports the understanding of grammar rules
- c. It assists in analysing sentence structure
- d. It aids in constructing well-formed sentences and paragraphs
- e. It helps in hardware optimisation

参考答案： ☒ b, ☒ c, ☒ d

解析： syntax 关注语言结构与语法规则，有助于分析句子结构并生成/识别 well-formed text。

Question 28

Which Pandas function is used to read a CSV file?

- a. `pd.read_csv()`
- b. `pd.to_csv()`
- c. `pd.read_excel()`
- d. `pd.load_csv()`

参考答案： ☒ a

解析： 读取 CSV 用 `pd.read_csv()`。

Question 32

Which parameter in `pd.read_csv()` specifies the delimiter?

- a. `delim`
- b. `split`
- c. `delimiter`
- d. `sep`

参考答案: `d`

解析: `pd.read_csv()` 指定分隔符最常用的是 `sep`。

Question 35

What does 'Character Data' consist of?

- a. Sequences of characters
- b. Single alphabetic letters, symbols, or numerals
- c. Quantifiable information
- d. Binary data

参考答案: `b`

解析: character data 指单个字符层面的数据, 如字母、数字、符号。

Question 37

Which technique is used to handle categorical data by creating a binary column for each category?

- a. One-hot Encoding
- b. Label Encoding
- c. Normalisation
- d. Vectorisation

参考答案: `a`

解析: 为每个类别开一列 0/1 特征, 就是 one-hot encoding。

Question 40

In the context of text data pre-processing, what is the purpose of case normalisation?

- a. To differentiate between common nouns and proper nouns
- b. To convert all the words into either uppercase or lowercase
- c. To increase the complexity of the text
- d. To identify and correct grammatical errors

参考答案: `b`

解析: case normalisation 就是把文本统一成大写或小写, 减少同词不同写法带来的碎片化。

Question 42

Exploratory Data Analysis (EDA) is used for:

- a. Finalizing data collection methods
- b. Reducing the volume of data
- c. Uncovering patterns, trends, and anomalies
- d. Reducing the volume of data

参考答案: ☒ c

解析: EDA 的目标是发现模式、趋势和异常, 不是定稿数据采集方案。

Question 55

Why is it important to remove stop words in text data pre-processing? (Select all that apply)

- a. To highlight more meaningful words in text analysis
- b. To prevent skew in analysis results
- c. To add more complexity to the model
- d. To increase data set size
- e. To reduce computational load

参考答案: ☒ a, ☒ b, ☒ e

解析: 去停用词是为了突出更有信息量的词、减少噪声并降低计算负担。

Question 57

To read an Excel file into a DataFrame, which function is used?

- a. `pd.load_excel()`
- b. `pd.read_excel()`
- c. `pd.read_csv()`
- d. `pd.import_excel()`

参考答案: ☒ b

解析: 读取 Excel 文件用 `pd.read_excel()`。

Question 63

How do you set a DataFrame's column 'ID' as its index?

- a. `df.index = 'ID'`
- b. `df.set_index('ID')`
- c. `df.set('ID')`
- d. `df.index_col('ID')`

参考答案: ☒ b

解析: 把某列设成索引常用 `df.set_index("ID")`。

Question 66

What does the `pd.read_json()` function do?

- a. Converts a DataFrame to a JSON object
- b. Writes data to a JSON file
- c. Reads data from a JSON file into a DataFrame
- d. Parses JSON data into a list

参考答案: ☒ c

解析: `pd.read_json()` 会把 JSON 读进 DataFrame。

Question 71

Which techniques are essential for handling numerical data in EDA? (Select all that apply)

- a. Descriptive statistics
- b. Encoding
- c. Tokenisation
- d. Vectorisation
- e. Normalisation

参考答案: ☒ a, ☒ e

解析: 数值型 EDA 常用 descriptive statistics, 也可能做 normalisation; encoding / tokenisation / vectorisation 不是数值数据核心方法。

Question 75

Which parameter in `read_csv()` is used to specify which rows should be used as header?

- a. top_row
- b. head_row
- c. header
- d. usecols

参考答案: ☒ c

解析: `header` 参数决定哪一行当列名。

Question 77

What does the `skiprows=3` parameter in `pd.read_csv()` do?

For example, `df = pd.read_csv('data.csv', skiprow = 3)`

- a. Skips any three rows randomly
- b. Skips first three columns from the left of the file
- c. Skips last three rows at the end of the file
- d. Skips first three rows from the beginning of the file

参考答案: ☒ d

解析: `skiprows=3` 表示从文件开头跳过前三行。

Question 81

What is the middle value after all values are ordered in a data set called?

- a. Mean
- b. Mode
- c. Variance
- d. Median

参考答案: ☒ d

解析: 有序数据中间那个值叫 median。

Question 88

Which of the following is crucial for preparing text for NLP tasks?

- a. Variance Calculation
- b. Tokenization
- c. Normalization
- d. Image Encoding

参考答案: ☒ b

解析: NLP 的第一步通常是 tokenization。

Question 90

Which measure of central tendency is the arithmetic average of a set of values?

- a. Median
- b. Mode
- c. Mean
- d. Range

参考答案: ☒ c

解析: 算术平均数就是 mean。

Question 96

For what reasons might case normalisation not always be needed in text data pre-processing? (Select all that apply)

- a. In information retrieval tasks
- b. In audio data processing
- c. For graphical data representation
- d. When performing named entity recognition
- e. To maintain the original meaning of words

参考答案: ☒ a, ☒ d, ☒ e

解析: 和 Q23/Q24 同理, IR、NER、保留原义等场景不一定要统一大小写。

Question 105

Which technique is used to handle categorical data by creating a binary column for each category?

- a. Label Encoding
- b. Vectorisation
- c. One-hot Encoding
- d. Normalisation

参考答案: ☒ c

解析: 和 Q37 同义, 这里正确项是顺序不同后的 one-hot encoding。

Question 110

Which data types are primarily analysed during EDA? (Select all that apply)

- a. Audio Data
- b. Categorical Data
- c. Numerical Data
- d. Text Data
- e. Image Data

参考答案: ☒ b, ☒ c, ☒ d

解析: 课程中的 EDA 主要看 categorical、numerical、text 三类。

Question 111

Which steps are involved in Lemmatisation for text data pre-processing? (Select all that apply)

- a. Reduction of words to their stems
- b. Application of morphological analysis
- c. Conversion of words to their base or dictionary form
- d. Use of part-of-speech tags
- e. Use of context surrounding the words

参考答案: `b, c, d, e`

解析: lemmatisation 依赖词形分析、词性和上下文, 把词还原成 dictionary form; “stem” 不是它的核心定义。

Question 120

Which method is used to get a summary of a DataFrame's structure and column types?

- a. `df.overview()`
- b. `df.summary()`
- c. `df.describe()`
- d. `df.info()`

参考答案: `d`

解析: 和 Q16 同义, 结构摘要用 `df.info()`。

Question 122

Which parameter in `pd.read_csv()` specifies the delimiter?

- a. `sep`
- b. `delim`
- c. `delimiter`
- d. `split`

参考答案: `a`

解析: 和 Q32 同义, 这里正确项换成了 `sep`。

Question 123

What is the aim of using Descriptive Statistics in EDA for Numerical Data?

- a. To manipulate the dataset
- b. To encode textual information into numerical values
- c. To provide insights into the central tendency, dispersion, and distribution shape
- d. To visualize data in 3D

参考答案: `c`

解析: descriptive statistics 的目标是概括中心位置、离散程度和分布形态。

Question 124

What does continuous data represent?

- a. Visual information encoded digitally
- b. Quantifiable information that can be divided infinitely
- c. Single alphabetic letters or symbols
- d. Countable values or categories

参考答案: **b**

解析: continuous data 是可以无限细分的可量化数值。

Question 131

Exploratory Data Analysis (EDA) is used for:

- a. Reducing the volume of data
- b. Uncovering patterns, trends, and anomalies
- c. Reducing the volume of data
- d. Finalizing data collection methods

参考答案: **b**

解析: 和 Q42 同义, EDA 用来发现 patterns / trends / anomalies。

Question 135

Which data visualisation method is primarily used for numerical data to understand its spread and central tendency?

- a. Pie Charts
- b. Histograms
- c. Mind Maps
- d. Flowcharts

参考答案: **b**

解析: 看数值分布与中心位置最常用 histogram。

Question 145

Which of the following are benefits of correctly identifying data types during EDA? (Select that all apply)

- a. Guides data cleaning and preprocessing
- b. Enables effective visualisation
- c. Determines the color scheme of visualisations
- d. Assists in feature engineering
- e. Allows for appropriate analysis

参考答案: **a, b, d, e**

解析: 正确识别数据类型能指导清洗、可视化、特征工程和后续分析。

Question 149

What challenges are presented by textual content in EDA for character and text data? (Select all that apply)

- a. Variability in text formats
- b. Data sparsity
- c. Ambiguity in language
- d. High dimensionality
- e. High computational costs

参考答案: ☒ a, ☒ b, ☒ c, ☒ d, ☒ e

解析: 和 Q2 同义, 文本常见难点包括格式多变、稀疏、高维、歧义和计算成本。

Question 151

What are the steps involved in Count Vectorisation? (Select all that apply)

- a. Vocabulary Building
- b. Normalisation
- c. Count Calculation
- d. Encoding
- e. Tokenisation

参考答案: ☒ a, ☒ c, ☒ e

解析: count vectorisation 通常先 tokenise、建立词表, 再统计每个词出现次数。

Question 153

Which of the following are goals of Exploratory Data Analysis (EDA)? (Select all that apply)

- a. To understand the patterns, anomalies, and relationships in the data
- b. To explore and summarise the main characteristics of the data visually
- c. To use statistical graphics, plots, and information tables
- d. To perform complex predictive modeling
- e. To manipulate the data set

参考答案: ☒ a, ☒ b, ☒ c

解析: EDA 目标是理解数据并做可视化总结, 不是直接做复杂预测建模。

Question 158

For what reasons might case normalisation not always be needed in text data pre-processing? (Select all that apply)

- a. In information retrieval tasks
- b. When performing named entity recognition
- c. For graphical data representation
- d. In audio data processing
- e. To maintain the original meaning of words

参考答案: ☐ a, ☐ b, ☐ e

解析: 和 Q23/Q24/Q96 同理, IR、NER、保留原始含义时不一定要大小写归一化。

Question 163

What is the aim of using Descriptive Statistics in EDA for Numerical Data?

- a. To provide insights into the central tendency, dispersion, and distribution shape
- b. To visualize data in 3D
- c. To encode textual information into numerical values
- d. To manipulate the dataset

参考答案: ☐ a

解析: 和 Q123 同义, 这里正确项换成了 ☐ a。

Question 164

What is the significance of identifying data types during EDA?

- a. It changes the textual content into images
- b. It influences how data will be processed and analyzed
- c. It is only important for numerical data
- d. It determines the type of cake to bake

参考答案: ☐ b

解析: 识别数据类型会直接影响后续怎么处理和分析数据。

Week 5 (25 题)

- 范围: Data Discovery & Collection

Question 10

What aspects does Data Cataloging and Metadata Management cover? (Select all that apply)

- a. Usage metadata
- b. Data anonymisation
- c. Metadata collection
- d. Data lineage documentation
- e. Catalog creation

参考答案: ☐ a, ☐ c, ☐ d, ☐ e

解析： cataloging / metadata management 关注 metadata collection、lineage、catalog 本身，也会包含 usage metadata；anonymisation 属于隐私保护，不是 cataloging 核心内容。

Question 31

Qualitative Data is:

- a. Exclusively about quantities
- b. Only numerical
- c. Descriptive and conceptual
- d. Limited to structured research instruments

参考答案： ☒ c

解析： qualitative data 强调描述性、概念性，不是纯数量。

Question 44

Data Cataloging aids in:

- a. Making it harder to find and understand data
- b. Ignoring data lineage
- c. Decreasing data accessibility
- d. Creating a searchable catalog of data assets

参考答案： ☒ d

解析： data catalog 的作用是让数据资产可搜索、可理解、可发现。

Question 49

Which is NOT a method of collecting structured data?

- a. Relational databases
- b. Online forms
- c. Surveys and questionnaires
- d. Text mining and NLP

参考答案： ☒ d

解析： text mining / NLP 主要处理非结构化文本，不属于 structured data collection 方法。

Question 50

Which of the following is NOT a part of the Data Discovery Process?

- a. Data Cataloging and Metadata Management
- b. Data Exploration and Visualization
- c. Data Encryption
- d. Data Integration and Consolidation

参考答案: ☒ c

解析: data encryption 属于安全控制, 不是 data discovery 流程本身。

Question 68

Factors to consider during the data collection phase include: (Select all that apply)

- a. Collection method efficiency
- b. Source reliability
- c. Data format
- d. Cost of data storage physical locations

参考答案: ☒ a, ☒ b, ☒ c

解析: collection 阶段常看方法效率、来源可靠性、数据格式; “physical locations” 不是这里的核心。

Question 72

Which are considered primary sources of data? (Select all that apply)

- a. Observations
- b. Experiments
- c. Industry reports
- d. Surveys
- e. Government publications

参考答案: ☒ a, ☒ b, ☒ d

解析: primary source 是直接采集的数据, 如 observations、experiments、surveys; industry reports / government publications 更像二手资料。

Question 84

Which is NOT a method of collecting structured data?

- a. Surveys and questionnaires
- b. Online forms
- c. Relational databases
- d. Text mining and NLP

参考答案: ☒ d

解析: 和 Q49 同理, text mining / NLP 不是 structured collection method。

Question 85

Which phase comes immediately after data discovery?

- a. Data storage
- b. Data collection
- c. Data cleaning
- d. Data analysis

参考答案: ☒ b

解析: discovery 后面紧接着就是 collection。

Question 89

Data Security measures are essential for:

- a. Ensuring data is openly accessible
- b. Secure storage and transmission of data
- c. Reducing data quality
- d. Encouraging unauthorized data access

参考答案: ☒ b

解析: data security 关注安全存储与传输，不是开放访问。

Question 91

Data Cataloging aids in:

- a. Creating a searchable catalog of data assets
- b. Ignoring data lineage
- c. Making it harder to find and understand data
- d. Decreasing data accessibility

参考答案: ☒ a

解析: 这题和 Q44 同义, catalog 的目的就是形成可搜索的数据资产目录。

Question 98

What roles does data play in decision-making? (Select all that apply)

- a. Crisis prediction and response
- b. Compliance monitoring
- c. Understanding customer behaviour
- d. Enhancing product quality
- e. Operational efficiency

参考答案: a, b, c, d, e

解析: 数据在决策里能支持风险、合规、客户洞察、产品和运营优化, 所以五项都对。

Question 107

Ethical considerations in Data Collection include: (Select all that apply)

- a. Privacy and anonymity
- b. Informed consent
- c. Data retention and disposal
- d. Data beautification
- e. Compliance with laws and regulations

参考答案: a, b, c, e

解析: 伦理采集通常包括隐私、知情同意、保留/销毁规范和法律合规。

Question 127

Qualitative Data is:

- a. Descriptive and conceptual
- b. Only numerical
- c. Limited to structured research instruments
- d. Exclusively about quantities

参考答案: a

解析: 和 Q31 同义, qualitative data 是描述性/概念性数据。

Question 128

The main advantage of using existing data is:

- a. Always perfectly matching specific needs
- b. Time efficiency and cost reduction
- c. Increased costs
- d. Reduced access to diverse data

参考答案: b

解析: existing data 的主要优势通常是省时省钱。

Question 132

Metadata in Data Cataloging includes:

- a. Only structural metadata
- b. Descriptive, Structural, and Technical Metadata
- c. Only usage metadata
- d. Only descriptive metadata

参考答案: ☒ b

解析: metadata 不只是一种类型, 常见有 descriptive、structural、technical 等。

Question 133

What role does data play in modern business decision-making?

- a. It is not significant in decision-making
- b. It is exclusively used for financial analysis
- c. It is only used for data storage
- d. It aids in understanding customer behavior and evidence-based strategic planning

参考答案: ☒ d

解析: 现代业务决策里, 数据能支持客户理解和基于证据的规划。

Question 134

Metadata Collection is critical for understanding:

- a. Only the data's numerical value
- b. Restricting data access
- c. How to delete the data
- d. The data's origin, format, and characteristics

参考答案: ☒ d

解析: metadata collection 帮助理解数据从哪来、什么格式、有哪些特征。

Question 136

Ensuring data privacy and security is crucial due to:

- a. Regulations
- b. Lack of regulations
- c. Unimportance of data sensitivity
- d. The ease of data handling

参考答案: ☒ a

解析: 隐私和安全之所以重要, 一个直接原因就是法规要求。

Question 137

Which of the following is NOT a part of the Data Discovery Process?

- a. Data Cataloging and Metadata Management
- b. Data Encryption
- c. Data Exploration and Visualization
- d. Data Integration and Consolidation

参考答案: ☒ b

解析: 这题和 Q50 同义, 只是正确项换成了 ☒ b。

Question 139

Text Mining and Natural Language Processing (NLP) are methods for collecting:

- a. Structured data
- b. Numerical data only
- c. Semi-structured data
- d. Unstructured data

参考答案: ☒ d

解析: text mining / NLP 处理的是 unstructured text。

Question 154

What are the roles of Data in modern business decision-making? (Select all that apply)

- a. Risk Management
- b. Enhancing product quality
- c. Compliance monitoring
- d. Understanding customer behaviour
- e. Performance Optimisation

参考答案: ☒ a, ☒ b, ☒ c, ☒ d, ☒ e

解析: 这题列出的 risk、quality、compliance、customer、performance 都是数据支撑决策的典型作用。

Question 159

Data Security measures are essential for:

- a. Encouraging unauthorized data access
- b. Reducing data quality
- c. Ensuring data is openly accessible
- d. Secure storage and transmission of data

参考答案: ☒ d

解析: 和 Q89 同义, 正确项换成了 ☒ d。

Question 161

Data Privacy measures are intended to:

- a. Expose sensitive information
- b. Increase data accessibility to the public
- c. Limit the use of data for analysis
- d. Protect sensitive information in compliance with regulations

参考答案: ☐ d

解析: privacy measures 的目标是依法保护敏感信息, 而不是公开它。

Question 162

Metadata Collection is critical for understanding:

- a. Only the data's numerical value
- b. How to delete the data
- c. Restricting data access
- d. The data's origin, format, and characteristics

参考答案: ☐ d

解析: 和 Q134 同义, metadata 帮你理解 origin / format / characteristics。

Week 6 (5 题)

- 范围: Data Structuring

Question 38

Graph data models are ideal for:

- a. Representing relationships between entities
- b. Ignoring entity relationships
- c. Simple numerical analysis
- d. Storing textual data only

参考答案: ☐ a

解析: graph data model 最适合表达实体之间的关系。

Question 51

What does Structured Data refer to?

- a. Data that is organized and easily understandable by machine language
- b. Multimedia data
- c. Social media content
- d. Data that is disorganized and random

参考答案: ☒ a

解析: structured data 是机器容易理解、字段组织清晰的数据。

Question 60

Data Profiling and Assessment involve understanding which types of data structure? (Select all that apply)

- a. Big data
- b. Time-series data
- c. Structured data
- d. Unstructured data
- e. Graph data

参考答案: ☒ b, ☒ c, ☒ d, ☒ e

解析: profiling / assessment 需要先看数据是什么结构, 因此 time-series、structured、unstructured、graph 都要识别。

Question 65

Which data structures are essential for Data Discovery and Collection? (Select all that apply)

- a. Relational databases
- b. Multimedia data
- c. Graph data models
- d. Time-series data
- e. Data warehouses

参考答案: ☒ a, ☒ b, ☒ c, ☒ d, ☒ e

解析: 这题把“数据结构”和“数据来源/类型”混在一起了; 按课程复习时先把这几类都记住。

Question 140

Time-series data is indexed in:

- a. Time order
- b. Random order
- c. Hierarchical order
- d. Alphabetical order

参考答案: ☒ a

解析: time-series 数据按时间顺序索引。
